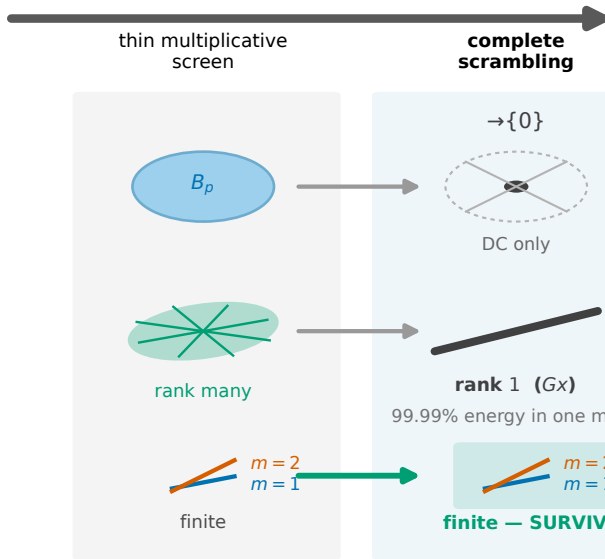


OPTICAL DISORDER



complete-scrambling law

$$\text{Cov}(b) = Q(x) (O \circ O) + R$$

$O \circ O$: code overlaps; R : noise floor

$$Q(x) = x^T G x, \quad G > 0$$

two directional classes:

$$x^T G \delta \neq 0 \Rightarrow m = 1$$

first-order observable

$$x^T G \delta = 0 \Rightarrow m = 2$$

curvature-rescued ($\delta^T G \delta > 0$)

$\Rightarrow$ **blind set empty**



amplitude anchor required
(else quotient info $j \rightarrow 0$)

Strong disorder erases **coordinates** (support $\rightarrow \{0\}$, rank $\rightarrow 1$) but not **local change observability**: the local change order m stays finite.